

ANURAG BADDUR

CAD/CAM • Application Engineer

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PROFESSIONAL SUMMARY

CAD/CAM Application Engineer with hands-on tool-room experience supporting VISI and ESPRIT EDGE for CNC/VMC and Swiss-type programming at a Hexagon reseller. Combines practical knowledge of die-casting tools, mould assembly, press tools, electrode design, wire-cut programming and reverse engineering (STL-to-CAD) with strong customer-facing skills. Diagnoses the real manufacturing requirement first, then builds the right software workflow around it. Builds plugins and customizations (AI-assisted, C#/.NET) that turn frustrated users into confident operators.

CORE COMPETENCIES

CAD/CAM Applications VISI, ESPRIT EDGE (20xx), NX, SolidWorks

Machining & Programming CNC & VMC Programming, Swiss-type Machine Programming

Tooling Expertise Press Tools, Mould Design & Assembly, Electrode Design, Wire-Cut Programming **Reverse Engineering**

STL / Scan Data to CAD Model Reconstruction (VISI), Surface & Solid Modelling from **Customization & Scripting** Mesh

Plugin Development (AI-assisted), C# and .NET (working knowledge)

Customer Engagement Requirement Analysis, Technical Support, User Training, Software Adoption

PROFESSIONAL EXPERIENCE

Application Engineer | Vihaans Technocraft (Hexagon Reseller), Pune Dec 2024 – Present • Support customers using VISI and ESPRIT EDGE for CNC/VMC programming — translate manufacturing requirements into practical, production-ready programming workflows.

~~• Handle Swiss-type machine programming end-to-end: guide customers through CAM setup, machine-specific settings, and deliver usable CNC~~

~~programs.~~•Handle mould and press-tool design processes and train customers across India on complete design-to-manufacture workflows.

• Perform reverse engineering from STL / scan data to accurate 3D CAD models in VISI — reconstruct surfaces and solids from mesh data to support mould, electrode and tooling design.

• Convert scanned STL data into production-ready models for electrode design, mould assembly and downstream machining preparation using VISI's reverse-engineering tools.

• Understood customer requirements on electrode design and automated key steps; also supported multiple customers on wire-cut programming with VISI.

• Support press-tool, mould and electrode design workflows alongside wire-cut programming, bridging design intent with machining preparation. •

Deliver hands-on technical support and structured training that converts struggling users into confident, self-sufficient operators.

• Build plugins and customizations with AI-assisted development (C# / .NET) to deliver precise, tailored solutions matched to customer requirements.

Tool Room Engineer | Terminal Technology, Pune Oct 2023 – Nov 2024 • Executed mould assembly and tool-room operations, collaborating daily with toolmakers and engineers on component fitting, alignment and final assembly.

• Supported machining, fitting and try-out activities for moulds and tools; assisted in tool maintenance and production handover.

• Gained hands-on exposure to machining, tooling and maintenance that now informs accurate diagnosis of manufacturing context behind every customer support case.

Tool Room Intern | Prabha Industries, Pune Jul 2022 – Sep 2023 • Assisted toolmakers with die-casting tool assembly, component fitting and alignment of core and cavity parts; supported electrode preparation and basic wire-cut tasks. • Built foundational understanding of die-casting tool construction, component function and standard tool-room practices.

NOTABLE PROBLEMS SOLVED & SOLUTIONS DELIVERED

• **Mould design & assembly workflows** — guided customers through complete mould design in VISI (core, cavity, slides, ejectors) and linked design output to machining and assembly checks.

• **Press-tool design support** — supported progressive and compound press-tool design processes, from strip layout and punch/die detailing through to production-ready tool models.

• **Electrode design automation** — automated electrode extraction, stock sizing and identification engraving so tool-room electrode jobs moved from concept to ready-to-machine models faster.

• STL-to-CAD reverse engineering — reconstructed accurate surfaces and solids from scan / mesh data in VISI so mould, electrode and tooling design could proceed from physical parts or legacy scans.

• **Design-to-machining bridge** — linked press-tool, mould and electrode design intent with CAM preparation and unblocked wire-cut programming end-to-end.

• **4-axis & 5-axis simultaneous programming** — set up and delivered multi-axis simultaneous toolpaths for complex components, reducing cycle time and improving surface finish.

• **Double-turret / double-spindle programming** — guided Swiss-type and multi-channel CNC setups (dual turret, dual spindle) end-to-end, synchronising operations into collision-free, production-ready programs.

• **Complex VMC / CNC workflow unblocking** — resolved machine-specific CAM settings and fixture constraints so customers could program and machine without repeated support loops.

• **User adoption turnaround** — turned at-risk users ready to abandon VISI / ESPRIT EDGE into confident designers and programmers through structured hands-on training.

• **Custom plugins (stock, engraving, PDF-to-3D)** — built AI-assisted C#/.NET customizations that automated electrode blank sizing, identification engravings and PDF-drawing-to-3D conversion.

EDUCATION

Diploma in Tool & Die Making — Government Tool Room & Training Center (GT&TC), 2019 – 2023 Dharwad

PCMB (Science) — Padmavati PU College, Hosur 2017 – 2019

LANGUAGES & INTERESTS

Languages: English, Hindi, Kannada | Interests: Chess, Football